Vectors I Homework 2: Scalar Product

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Scalar product: properties, angles, perpendicularity, projections. Q1 True/False from the Basics Worksheet (Scalar Product). Q2–Q6 are full past-paper questions.

1 For the following statements, state whether it is **true** or **false**.

	Statement	**T/F**
(a)	$\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$.	
(b)	$\mathbf{a}\cdot\mathbf{a}=\lvert\mathbf{a}\rvert^2$.	
(c)	$\mathbf{i}\cdot\mathbf{j}=0$.	
(d)	$\mathbf{a}\cdot\mathbf{b}$ is a vector.	
(e)	$\mathbf{i}\cdot\mathbf{j}=\mathbf{k}$.	
(f)	If $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$, then $\mathbf{a}\cdot\mathbf{b}=0$.	
(g)	$(2\mathbf{a})\cdot(3\mathbf{b})=6\,\mathbf{a}\cdot\mathbf{b}$.	
(h)	$(\mathbf{a}\cdot\mathbf{b})\mathbf{c}$ is a vector.	
(i)	$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$, where θ is the non-reflex angle between $\mathbf{a}$ and $\mathbf{b}$.	
(j)	$\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}=\begin{pmatrix}2\\6\\15\end{pmatrix}$.	
(k)	If $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$.	
(l)	$\lvert\mathbf{a}\cdot\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$.	
(m)	$\mathbf{a}\cdot\mathbf{b}\le\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$.	
(n)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}>0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is acute.	
(o)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}<0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is obtuse.	
(p)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}\ge 0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is acute.	
(q)	$\mathbf{a}\cdot(\mathbf{b}\cdot\mathbf{c})$ is a scalar.	
(r)	$(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b})=\lvert\mathbf{a}\rvert^2+\lvert\mathbf{b}\rvert^2$.	
(s)	For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$, then $\mathbf{a}$ and $\mathbf{b}$ are parallel.	
(t)	$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$ also holds if θ is the reflex angle between $\mathbf{a}$ and $\mathbf{b}$.	
(u)	Let θ be the angle ABC. Then $\overrightarrow{BA}\cdot\overrightarrow{BC}=\lvert\overrightarrow{BA}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$.	
(v)	The length of projection of $\mathbf{a}$ onto $\mathbf{b}$ is equal to the length of projection of $\mathbf{b}$ onto $\mathbf{a}$.	
(w)	If $\lvert\mathbf{a}\rvert=\lvert\mathbf{b}\rvert$, then $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=0$.	
(x)	$\mathbf{a}\cdot\mathbf{b}=\mathbf{a}\cdot\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$.	

	Statement	**T/F**
(y)	Let θ be the angle ABC. Then $\overrightarrow{AB}\cdot\overrightarrow{BC}=\lvert\overrightarrow{AB}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$.	
(z)	Let θ be the angle CBD. Then $\overrightarrow{BD}\cdot\overrightarrow{CB}=\lvert\overrightarrow{BD}\rvert\lvert\overrightarrow{CB}\rvert\cos\theta$.	

2 [TJC 2025 Prelim P1 Q6]

The points A and B lie on a circle with centre O and radius λ units. With reference to the origin O, points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. The point X on the line segment AB is such that $AX:XB=1:3$ and the point Y is the foot of the perpendicular from X to OB.

(a) Find the position vector of X. [1]

(b) It is given that the acute angle AOB is $\frac{\pi}{6}$. Find $\mathbf{a}\cdot\mathbf{b}$ in terms of λ. [1]

(c) Show that the position vector of Y is $\frac{2+3\sqrt{3}}{8}\mathbf{b}$. [2]

(d) Hence find the exact area of ΔOXY in terms of λ. [4]

3 [HCI 2025 Prelim P2 Q4]

Let A, B and C be the points on the same plane with position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively and $\mathbf{a}+\mathbf{b}+\mathbf{c}=\mathbf{0}$. It is given that vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are unit vectors.

(a) (i) By considering $\mathbf{c}\cdot\mathbf{c}$, find the value of $\mathbf{a}\cdot\mathbf{b}$. [3]

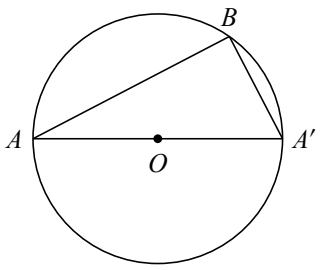
(ii) Find the angle $\angle AOB$. [2]

(iii) Draw the position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ on a single diagram. Using your diagram, identify the type of triangle ABC. [2]

(b) The point D has position vector $\mathbf{a}+\mathbf{b}$. Find the area of the quadrilateral $ACBD$. [2]

4 [SAJC 2025 Prelim P1 Q11]

The points A, B and A' lie on a circle with centre O such that $\overrightarrow{OA}=\mathbf{a}$ and $\overrightarrow{OB}=\mathbf{b}$. It is also given that the line segment AA' is a straight line that passes through O (see diagram).



(i) Using a suitable scalar product, prove that $\angle ABA'=90°$. [4]

(ii) Express the area of $\Delta ABA'$ in the form $k|\mathbf{a}\times\mathbf{b}|$, where k is a real constant. [2]

5 [VJC 2025 Prelim P1 Q3]

Referred to the origin O, points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. Point C lies on OA, such that $OC:CA=2:1$. Point D lies on OB, such that $OD:DB=\lambda:\mu$. It is given that the area of triangle ABD is half the area of triangle ABC.

(a) Show that the area of triangle ABD is given by $\frac{\mu}{2(\lambda+\mu)}|\mathbf{a}\times\mathbf{b}|$. Hence find the ratio $\lambda:\mu$. [4]

(b) The point E has position vector $\frac{1}{4}\mathbf{a}+\frac{5}{8}\mathbf{b}$. Show that A, E and D are collinear. [3]

It is further given that the angle AOB is $\frac{\pi}{4}$ and O lies on the perpendicular bisector of the line segment AB.

(c) Find the length of projection of $\mathbf{a}$ on $\mathbf{b}$, giving your answer in terms of $|\mathbf{b}|$. Hence find the position vector of the point F, the foot of the perpendicular from A to OB. [3]

6 [CJC 2024 Prelim P2 Q1]

(a) The points P, Q and R have position vectors $\mathbf{j}+\mathbf{k}$, $\mathbf{i}-2\mathbf{j}+3\mathbf{k}$ and $-2\mathbf{i}+\mathbf{j}+4\mathbf{k}$ respectively. Find the exact length of projection of $\overrightarrow{QR}$ on $\overrightarrow{PQ}$. [3]

(b) Two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ are such that $|\mathbf{a}|=\lambda|\mathbf{b}|$ and $|\mathbf{a}+\lambda\mathbf{b}|=2|\mathbf{a}-\lambda\mathbf{b}|$, where λ is a positive constant. Using a suitable scalar product, show that $\cos\theta=\frac{3}{5}$, where θ is the acute angle between $\mathbf{a}$ and $\mathbf{b}$. [3]

END OF HOMEWORK